

Paging Advantages

- ✓ Easy to allocate memory
 - Memory comes from a free list of fixed size chunks
 - Allocating a page is just removing it from the list
 - External fragmentation not a problem
- ✓ Easy to swap out chunks of a program
 - All chunks are the same size
 - Use valid bit to detect references to swapped pages
 - Pages are a convenient multiple of the disk block size

Paging Limitations

- Can still have internal fragmentation
- Requires 2 or more references, which could limit performance
- ➔ **Solution:** use a hardware cache of lookups (coming next)
- The amount of memory to store the page table is significant
 - Need one PTE per page, with 32 bit address space w/ 4KB pages = 2^{20} PTEs
 - 4 bytes/PTE = 4MB/page table
 - 25 processes = 100MB just for page tables!
- ➔ **Solution** : page the page tables (coming next)